Metric Scale in Image-to-3D Generation Is Delegated to the Depth Anchor

Anonymous ECCV 2026 Submission

Paper ID #*****

Abstract. Image-to-3D generative models output objects in a canonical unit cube, seemingly blind to physical size. We show this framing is wrong for depth-conditioned generators: the generator predicts a *dimensionless* object–scene scale ratio, and metric scale is *delegated* to the conditioning pointmap. On the same frozen SAM 3D Objects model, substituting a ground-truth metric pointmap for the default affine-invariant one drops isotropic object-size error from 71.5% to 6.7% — with no new parameters. Guided by this analysis, we recover metric object dimensions from a *single RGB image*: we swap the anchor to a monocular metric estimator (MoGe-2) and lightly fine-tune the layout pathway, reaching 17.6% mean per-axis dimension error across three real-world sources and 73.1 mAP@IoU$_{0.25}$ on NOCS-Real275 — surpassing RGB-only baselines and, on clean data, exceeding the accuracy of the raw anchor itself. Unlike prior metric layout evaluations, no ground-truth depth is used at training or inference.

Keywords: Metric scale · Image-to-3D generation · Monocular geometry

1 Introduction

Image-to-3D generative models have reached a level of visual fidelity where a single photograph is sufficient to produce a detailed mesh or Gaussian splat of an object with plausible geometry and texture. Foundations such as SAM 3D Objects [8] and TRELLIS [13] train on millions of synthetic and real objects, learning rich priors over 3D shape and surface appearance. Yet their outputs live in a canonical unit cube, $[-0.5, 0.5]^3$: a reconstructed bowl and a reconstructed car are indistinguishable in scale. Without absolute scale, a 3D generative model is an artist's tool, not a spatial computing primitive — robotic grasping, AR object placement, and multi-object scene reconstruction all require physical units.

The standard reading of this limitation is that the *model* lacks metric awareness, and that recovering scale requires either category-size priors (which plateau at the within-category variance), a separate metric-depth network bolted on after the fact, or expensive backbone retraining. This paper argues for a different reading.

Metric scale is delegated, not absent. Depth-conditioned image-to-3D pipelines condition generation on a pointmap — a dense per-pixel field of (x, y, z) coordinates — and normalise the object pose into scale-shift-invariant (SSI) space using a robust scene statistic of that pointmap. We show (Sec. 3) that the pose decoder of SAM 3D Objects *already* inverts this normalisation at inference: predicted object size is the product of a dimensionless, learned object–scene ratio and the input pointmap's scene scale. The generator never learns metric scale because it never has to — scale is *delegated* to whatever depth anchor conditions the model. The default anchor (MoGe [10]) is deliberately *affine-invariant*, so the delegated scale is meaningless; this, not a missing capability, is why outputs appear non-metric.

Three controlled experiments substantiate and sharpen this claim, all on the same frozen model. **(i) Oracle:** feeding a ground-truth metric pointmap instead of the MoGe default reduces median isotropic size error on NOCS-Real275 from 71.5% to 6.7% — zero trained parameters (Sec. 3.2). **(ii) Error decomposition:** across three real-world sources, applying a perfect *isotropic* correction to the model's predicted dimensions removes nearly all cross-source error (per-axis MAPE drops to 7.7/14.4/22.5% on NOCS/Objectron/ARKitScenes); object *proportions* are already accurate, the single scalar anchor factor is the bottleneck (Sec. 3.3). **(iii) Transfer asymmetry:** on Hypersim, a domain never used to train any component, size read through the pointmap pathway achieves 32% error while a head that regresses absolute size from the (scale-free) latent collapses to over 200%, reverting to training-set size priors (Sec. 3.4). Together these show *where the metric signal lives*: anchor object size to the pointmap; do not re-regress it from representations that are scale-invariant by construction.

Exploiting delegation. The analysis prescribes a recipe (Sec. 4). First, swap the conditioning pointmap for a *monocular metric* estimator (MoGe-2 [11]); the pipeline is explicitly designed to consume estimated pointmaps, so this is an in-distribution change, and it alone improves mixed-source dimension error from 22.3% to 20.2%. Second, because a monocular anchor carries systematic, source- and size-dependent bias (*e.g.* it over-measures close-range objects by up to 1.76×) that the anchor alone cannot correct, we lightly fine-tune the model's layout pathway — a compact scale head, the spatial-latent cross-attention, and the layout branch of the Mixture-of-Transformers backbone (the shape branch stays frozen) — so the model learns a per-instance *correction to its own anchor*. This yields our best model: **17.6%** mean per-axis dimension error (10.3% median) across NOCS-Real275, Objectron, and ARKitScenes, approaching the oracle floor of the decomposition (∼15%). On NOCS-Real275, the fine-tuned model's isotropic size error (3.9% median) falls *below* the 6.7% the same frozen model achieves with a ground-truth depth pointmap — direct evidence that the model corrects anchor bias rather than merely inheriting the anchor. The same delegation mechanism supports full metric 3D bounding boxes: with predicted pose, we obtain 73.1 mAP@IoU$_{0.25}$ on NOCS-Real275 from RGB alone, surpassing RGB-only baselines (NOCSformer 43.5 [6], Cube R-CNN 14.9 [4]).

A practical consequence we highlight for metric scene understanding: prior metric layout evaluations of image-to-3D systems — including SAM 3D's own — use ground-truth depth as the metric reference at evaluation time. Our entire pipeline, training and inference, uses *no ground-truth depth*: the metric anchor is estimated from the same single RGB image. This is the setting in which such systems will actually be deployed.

Contributions.

- **A delegation analysis of metric scale** in depth-conditioned image-to-3D generation: the mechanism (Sec. 3.1), an oracle experiment isolating the depth source (71.5%→6.7% with zero trained parameters), and an error decomposition showing a single isotropic anchor factor dominates cross-source dimension error.
- **A transfer-asymmetry result**: pointmap-anchored size transfers to unseen domains; absolute-size regression from scale-free latents does not. This settles *where* metric supervision should attach.
- **A lightweight recovery recipe** — metric-anchor swap plus targeted SFT of the layout pathway — that needs no ground-truth depth, no new architecture, and no backbone retraining, evaluated as an ablation ladder isolating each ingredient.
- **Evidence the model corrects its anchor**: post-SFT accuracy on clean data exceeds the raw metric-anchor oracle, and full-box mAP corroborates that the recovered dimensions are correctly placed in 3D.

2 Related Work

Image-to-3D generative models. Large-scale image-to-3D systems have converged on a two-stage latent flow-matching design. TRELLIS [13] introduces the Structured 3D Latent (SLAT) representation, a sparse voxel grid carrying DINOv2 visual features; a rectified-flow transformer decodes SLAT tokens conditioned on a single image, with separate VAE heads producing Gaussian splats or textured meshes. TRELLIS v2 [12] replaces multiview-supervised features with native O-Voxel representations across four billion parameters. SAM 3D [8] extends the paradigm to natural images with clutter and occlusion: a 1.2B-parameter Mixture-of-Transformers (MoT) geometry model jointly estimates coarse shape and object layout (pose, translation, scale), followed by a sparse latent refinement model. All three systems represent objects in a normalised unit cube. Our starting observation is that in SAM 3D Objects the connection to physical units nonetheless exists — through the conditioning pointmap — and that understanding this delegation is the key to metric recovery.

Monocular geometry and metric depth estimation. ZoeDepth [3] fine-tunes a relative-depth backbone with per-domain metric bins; Metric3D [14] achieves scale-aware depth through canonical-camera normalisation. MoGe [10] reframes the problem as *affine-invariant* pointmap prediction, achieving state-of-the-art relative geometry while deliberately leaving global scale and shift unconstrained; MoGe-2 [11] restores metric scale via a decoupled global-scale head, motivated by the observation that entangling metric scale with relative geometry destabilises both — an observation our cross-dataset findings independently corroborate at the object level. MoGe is the depth front-end of the SAM 3D Objects pipeline we build on. Our work differs from the metric-depth literature in target and mechanism: we recover *object-level dimensions through a generative 3D reconstruction*, and we show the reconstruction model can learn to *correct* the systematic bias of its own monocular anchor — something the depth model alone cannot do.

Category-level size and pose estimation. NOCS [9] defines the normalised-object-coordinate-space framework for category-level 6-DoF pose and size; NOCS-Real275 is our cleanest evaluation source. Cube R-CNN [4] regresses metric 3D boxes directly from RGB across the unified Omni3D benchmark. OmniNOCS [6] unifies canonical 3D box annotations across ten sources (97 categories, 380k images) and provides NOCSformer, a class-agnostic 3D lifting model; we train and evaluate on its NOCS-Real275, Objectron [1], and ARKitScenes [2] subsets. These discriminative methods regress size against category priors learned in training — a strategy our own diagnostics show plateaus at the within-category size variance (an R^2 of only 0.35 between category and log-size on ARKitScenes). Our approach instead measures each instance against a per-image metric anchor, and uses learned priors only to correct the anchor's residual bias.

Adapting frozen generative models. Parameter-efficient adaptation such as LoRA [5] and ControlNet [15] injects new conditioning or behaviour into frozen backbones, typically for semantic control. We apply the same philosophy to a metric objective: all shape-generating weights stay frozen (SAM 3D's own ablations show shape quality does not depend on pointmap conditioning), and only the layout pathway — the part of the model that reads the anchor — is adapted.

3 Where Metric Scale Lives: A Delegation Analysis

3.1 Mechanism: the pose decoder delegates scale to the pointmap

SAM 3D Objects conditions its geometry model on a pointmap $\mathbf{P} \in \mathbb{R}^{H\times W\times 3}$ produced by MoGe [10] (or supplied externally, *e.g.* from a LiDAR sensor). The pipeline normalises the object into scale-shift-invariant (SSI) space using a robust scene statistic of the input pointmap,

$$s_{\text{scene}} = \text{mean}\big(|\mathbf{P} - \text{med}_z(\mathbf{P})|\big), \tag{1}$$

the mean absolute deviation about the median depth, computed *in the pointmap's own units*. The generator's layout branch predicts a per-axis SSI scale $\tilde{\mathbf{s}}$ — a dimensionless object–scene ratio — and the pose decoder lifts it back to world units at inference:

$$\mathbf{d}_{\text{pred}} \;=\; \mathbf{c} \odot \tilde{\mathbf{s}} \cdot s_{\text{scene}}, \tag{2}$$

Table 1: The depth source, not the model, controls metric accuracy. Median isotropic size error of the *frozen* pose-decoder pathway under three input pointmaps (same model, same images, same seeds; held-out instances: 64 NOCS / 163 Objectron / 148 ARKitScenes). GT pointmaps exist only for NOCS. Bias is the median ratio pred/GT.

Input pointmap	NOCS	Objectron	ARKitScenes	Overall	Bias (N/O/
MoGe (affine-invariant, default)	75.2	260.9	25.6	81.1	1.75 / 3.61 /
MoGe-2 (monocular metric)	16.7	75.7	19.6	29.2	1.16 / 1.76 /
GT depth (oracle)	**6.7**	—	—	—	≈1.0

where $\mathbf{c}$ is the canonical extent of the generated shape. Equation 2 is the delegation: *predicted object dimensions are directly proportional to the units of the input pointmap.* If the pointmap is metric, the output is metric; if the pointmap is affine-invariant — as MoGe's deliberately is — the output scale is arbitrary. The model has no metric knowledge of its own and needs none; it has a well-calibrated notion of *how large the object is relative to its scene.*

3.2 Oracle: the same frozen model is metric, given a metric anchor

We test the mechanism end-to-end on the frozen public model with *zero* trained parameters, varying only the input pointmap. On NOCS-Real275 (the one source with registered ground-truth depth and intrinsics), we compare the default MoGe anchor against a ground-truth metric pointmap built from sensor depth, and against MoGe-2 self-calibrated metric pointmaps as the deployable middle ground. Table 1 reports median isotropic size error ($|\max \mathbf{d}_{\text{pred}} - \max \mathbf{d}_{\text{GT}}| / \max \mathbf{d}_{\text{GT}}$) on held-out instances from all three sources.

Three observations. First, the ground-truth anchor achieves **6.7%** median error with no metric supervision anywhere in the model — confirming Eq. 2 carries the full metric signal. Second, the affine default is not merely noisy but *systematically biased* per source (up to 3.6× on Objectron), which is what any learned head trained on top of it must spend its capacity compensating. Third, the deployable monocular anchor (MoGe-2) recovers most of the oracle gap at room scale but retains a strong close-range bias (1.76× on Objectron) — a *learnable* residual, as its per-instance log-ratio scatter is small ($\sigma_{\log} \approx 0.09$ on NOCS).

3.3 Decomposition: one isotropic factor dominates dimension error

Predicted dimensions factor as proportions × isotropic scale (Eq. 2). To locate the error, we apply a per-instance *oracle isotropic correction* to the frozen model's predictions — rescaling each prediction by the single scalar that matches its true isotropic size, while keeping the predicted proportions untouched. Per-axis MAPE falls from 19/100/33% to **7.7** / **14.4** / **22.5%** (NOCS/Objectron/ARKitScen nearly all cross-source dimension error lives in the one scalar the pointmap anchors, while the generator's *proportions* are already accurate. This defines an oracle floor of roughly 15% overall for any method that fixes only the isotropic factor, and it concentrates the problem: correct the anchor's scalar bias, per instance.

3.4 Transfer asymmetry: anchor, don't regress

Should metric size be *regressed* from the model's latent features, or *anchored* to the pointmap? The latents are scale-invariant by construction (they were pretrained on canonically normalised shapes), so a regression head can only learn dataset-specific size priors. We test this on Hypersim [7] — a photorealistic indoor domain unseen by every component — using ground-truth depth pointmaps to remove anchor-quality confounds (n=400 instances). Size read through the pointmap pathway (Eq. 2) achieves **32.0%** MAPE overall and wins in every size bucket; a learned head that regresses absolute dimensions from pooled latent features (trained on NOCS/Objectron/ARKitScenes) collapses to 207.8%, reverting to its training-set size priors (869% on small objects). The head's number is out-of-distribution by construction — that is precisely the point: *the pointmap pathway transfers; absolute-size regression does not.* Consistent with this, we find that fine-tuning on ARKitScenes with the *affine* anchor yields metric gains that do not transfer to external benchmarks (HAMMER, iBims-1), while scale-invariant shape quality transfers unchanged — under a non-metric anchor, apparent metric learning is dataset memorisation.

Design consequences. The analysis dictates the method: (1) make the anchor metric (swap MoGe $\rightarrow$ MoGe-2); (2) attach supervision to the *dimensionless* SSI pathway so the pointmap keeps doing the metric lifting; (3) give the model the anchor's own statistics as features, so it can learn the per-instance *residual correction* that the raw anchor provably cannot apply to itself.

4 Method: Exploiting Delegation

4.1 Metric anchor swap

We replace the pipeline's MoGe front-end with MoGe-2 [11] full-frame metric pointmaps under self-estimated intrinsics (we verified that supplying ground-truth intrinsics *hurts* — forcing an external FOV fights the jointly-estimated depth/FOV — so no camera calibration is needed at inference). The pipeline is explicitly designed to consume estimated pointmaps, so this is an in-distribution substitution, not an architectural change: object-centric crops and the scene statistic of Eq. 1 are derived from the injected pointmap exactly as before.

4.2 Scale head and token injection

A compact *MetricScaleHead* (<0.2M parameters) summarises the metric evidence into a single conditioning token. Its input concatenates the pooled 8-dim sparse-structure (SS) shape latent, the SS generator's 3-dim log-SSI scale features, and the anchor statistics $\log s_{\text{scene}}$ and the object's median depth offset Δ_z — 13 dimensions total, all computed by the standard pipeline with no extra forward pass:

$$\mathbf{s} = \mathrm{MLP}\big([\bar{\mathbf{z}}_{\mathrm{SS}};\, \mathbf{f}_{\text{scale}};\, \log s_{\text{scene}};\, \Delta_z]\big) + \boldsymbol{e}_{\text{metric}}. \tag{3}$$

The token is appended to the spatial-latent (SLAT) decoder's cross-attention context via a transparent proxy around the condition embedder; a learned modality embedding $\boldsymbol{e}_{\text{metric}}$ distinguishes it from image tokens. A *MetricScaleDecoder* (<0.05M) pools the refined SLAT features (scatter-mean over active voxels) with a linear projection of the scale token and regresses $[\log W, \log H, \log D]$ in metres. During training, the injected token is dropped (zeroed) with probability 0.1, mirroring classifier-free-guidance dropout [13].

4.3 Targeted fine-tuning of the layout pathway

The SAM 3D geometry model is a Mixture-of-Transformers with parameter-disjoint shape and layout branches. We freeze the shape branch entirely — SAM 3D's own ablations show shape quality does not depend on pointmap conditioning, and we possess only 3D box labels, not ground-truth shape — and unfreeze, at discriminated learning rates:

- the SLAT cross-attention sublayers (24 blocks, ~101M parameters, lr 10^{-6} with 1,500-step linear warmup), which read the injected token;
- the MoT *layout* branch (per-block norms, self-attention, MLP; lr 10^{-5}), which produces the SSI scale the anchor lifts;
- the scale head and decoder (lr 10^{-4}).

Routing the two MoT branches by their module keys is essential: an earlier variant that unfroze the backbone uniformly trained only the tiny layout read-heads quickly while the layout transformer crawled, plateauing per-axis error ~10 points higher. Because the backbone runs in bfloat16 and the scale token is unbounded, all unfrozen cross-attention and norm sublayers are wrapped in an fp32 upcast shim — without it, training diverges with NaN attention weights within the first hundred steps. The objective is smooth-L1 on log-dimensions; the shape branch receives no gradient (verified: exactly zero gradient norm on all shape parameters). Total trained: ~500M of ~2B pipeline parameters, on a single A100 for 12 epochs over ~10k object records.

5 Experiments

5.1 Setup

Data. We train on the three real-world OmniNOCS [6] sources: NOCS-Real275 (tabletop, 6 categories), Objectron (hand-held captures), and ARKitScenes (room-

Table 2: Ablation ladder for metric dimension recovery. Same recipe, data, and held-out split throughout; each row changes one ingredient. Per-axis MAPE (%), mean over held-out instances; overall median in parentheses. MAE in cm.

Model	NOCS	Objectron	ARKitScenes	Overall	M
Category-mean prior (GT labels)†	13.5	22.2	43.6	—	
Head + injection, MoGe anchor (affine)	7.4	17.7	34.3	22.3 (11.9)	
+ metric anchor (MoGe-2)	7.9	17.4	28.6	20.2 (12.0)	
+ layout-branch SFT (full recipe)	**4.6**	**16.5**	**24.5**	**17.6** (**10.3**)	
Oracle iso-correction floor (Sec. 3.3)	7.7	14.4	22.5	∼15	

†Regression-derived approximation (residual RMSE in log space), reported per source; it uses ground-truth category labels, which our method does not.

scale furniture), capped at 3,334 records per source with balanced sampling (∼10k total). Held-out evaluation uses disjoint records: 64 NOCS / 200 Objectron / 200 ARKitScenes instances (fixed split, seed 0, identical across all rows of every table).

Metrics. Mean and median absolute percentage error (MAPE) of predicted vs. ground-truth per-axis metric dimensions $[W, H, D]$, plus per-source breakdowns; and official symmetry-aware NOCS 3D-box mAP at IoU 0.25/0.5 for full-pose corroboration (Sec. 5.4). We report both mean and median because the sources have heavy category tails (Sec. 5.5). The metric readout uses the same flow schedule as training (4 sparse-structure / 1 SLAT steps); this is faster than the stock generation schedule and we state it as part of the protocol, since the metric head does not transfer zero-shot to feature statistics from longer schedules.

Baselines. (1) *Category-mean prior*: regressing log-size on ground-truth category labels — the ceiling for any prior-based method — yields approximately 13.5/22.2/43.6% per source. Notably, adding object–camera distance to this regression improves it by at most 2 points, and category+distance explain only R^2=0.35 of ARKitScenes log-size variance: priors saturate, per-instance measurement is required. (2) *Depth-bridge*: measure the object directly from the masked MoGe-2 point cloud (percentile-robust bounding box). This uses the same monocular network as our anchor but no reconstruction model; it wins at room scale (22.7% median on ARKitScenes) and fails close-range (71.7% on Objectron), motivating a learned combination rather than either component alone.

5.2 Main result: the delegation ladder

Table 2 is the paper in one table: the same architecture, data, split, and schedule, changing one ingredient per row.

The anchor swap alone (row 2→3) improves overall error by 2.1 points, concentrated on room-scale ARKitScenes (34.3→28.6) — exactly where the affine anchor's bias was largest. Layout-branch SFT (row 3→4) contributes a further 2.6 points and brings every source within a few points of the decomposition's oracle floor. Most tellingly, on the matched isotropic-median statistic of Table 1, **the fine-tuned model reaches 3.9% on NOCS — below the 6.7% the same frozen model achieves with a ground-truth depth pointmap**: it is not merely inheriting its anchor but correcting the anchor's per-instance bias using appearance and shape context, which the depth network alone cannot do. The residual gap to the floor sits in Objectron (close-range anchor bias) and ARKitScenes category tails (Sec. 5.5).

Does the model correct the anchor, or just inherit it? We test this with a pre-registered analysis: on the same 422 held-out instances, same precomputed MoGe-2 pointmaps, and same seed, we compare the *raw anchor* (frozen pose-decoder size readout) against the fine-tuned model, regressing the log size-ratio log(pred/GT) on log object size. The raw anchor has a strong size-dependent bias — slope -0.38, over-measuring the two smallest size quartiles by $1.35\times$ and $1.70\times$ (median ratio) — which a global calibration constant cannot remove (44.6% MAPE after removing the best single constant). The fine-tuned model **flattens the slope to** -0.07 and reaches 16.7% MAPE — a 27.9-point improvement over the calibrated anchor (95% paired-bootstrap CI $[22.8, 33.5]$) — with per-source median ratios of 0.97/0.98/1.02 (NOCS/Objectron/ARKitScenes) versus the raw anchor's 1.16/1.76/0.91. Re-calibrating the fine-tuned model changes nothing (16.9%), *i.e.* its residual is no longer a systematic bias. The model has learned a per-instance, size-dependent correction that no constant — and hence no post-hoc calibration of the depth network — can express. This is the sense in which the contribution is a *model*, not a depth swap.

5.3 How should the model read the anchor? Head-representation ablation

With generators frozen (features cached), we compare four scale-decoder designs on identical inputs and splits: absolute log-dimension regression; the same plus explicit *anchor features* (the pose-decoder's delegated isotropic size, log pose scale, and voxel extents); a factored variant with gradient-decoupled isotropic/proportion branches (after MoGe-2 [11]'s decoupled-scale loss); and a binned-classification isotropic head (after [6]). Absolute regression plateaus at 35.4% mean and then overfits; adding anchor features improves it to **27.2%** (-8.2 points); factored (28.3%) and binned (27.6%, best median 14.4%) are statistically indistinguishable from the anchor-feature model. *Access to the anchor is first-order; the representation of the correction is second-order* — converging evidence for the delegation thesis from the opposite direction. (These cached-feature runs lack the live SLAT capacity of Table 2 and are mutually comparable only.)

5.4 Full 3D boxes: mAP corroboration

Metric dimensions are only useful if they are correctly placed. We evaluate full predicted 3D boxes (predicted rotation, translation, and size; ground-truth 2D

Table 3: 3D-box mAP on NOCS-Real275. Top: the depth-source A/B on a fixed trained model (8 frames): the anchor moves mAP from zero to competitive. Bottom: full-set evaluation (150 frames). RGB-only = no depth sensor input at inference. *Uses depth input at inference. Baselines from [6].

	Method / condition	mAP@0.25	mAP@0.5
Depth-source A/B (same trained model)	MoGe anchor (affine)	0.0	0.0
	MoGe-2 anchor	40.3	4.9
	GT-depth anchor (oracle)	89.8	17.6
Full set, 150 frames	Cube R-CNN [4] (RGB)	14.9	4.1
	NOCSformer [6] (RGB)	43.5	10.6
	NOCS [9] (RGB-D)*	79.6	72.4
	Ours, joint layout SFT (RGB)	**73.1**	8.9

instances) with the official symmetry-aware NOCS mAP on NOCS-Real275, RGB-only at inference.

The A/B (top of Table 3) is the delegation mechanism seen through mAP: with the affine anchor the boxes are metrically misplaced (53% translation error $\Rightarrow$ 0 mAP); the monocular metric anchor recovers 40.3; the GT anchor exceeds the *depth-supervised* NOCS baseline at IoU 0.25. Our jointly fine-tuned model reaches **73.1 mAP@0.25 from RGB alone** — 30 points above the best RGB-only baseline and approaching the RGB-D method — with median translation error 8.3 cm. mAP@0.5 remains bounded by that $\sim$8 cm translation precision, which our analysis attributes to the monocular anchor's depth accuracy, not to size (an oracle study shows size errors below 30% do not affect mAP@0.5 given correct pose).

The dimension model (Table 2) and the box model here are two fine-tunes of the same backbone: the mAP model co-trains the layout branch against pose+size targets, while the dimension model trains it only through the metric-head loss, which we find drifts the frozen pose read-heads' translation calibration ($\sim$0.5 m error). We therefore report each model on its own task and leave a single jointly-supervised checkpoint — warm-started from the dimension model — to future work.

5.5 Failure modes and limitations

We state these explicitly; each is a measured property of the system, not a conjecture.

The headline tracks the anchor. By design, accuracy inherits the anchor's failure modes. MoGe-2 over-measures close-range objects (1.76$\times$ median on Objectron); SFT reduces but does not eliminate this (Objectron 16.5% vs. floor 14.4%). Where the anchor collapses entirely, so will the system.

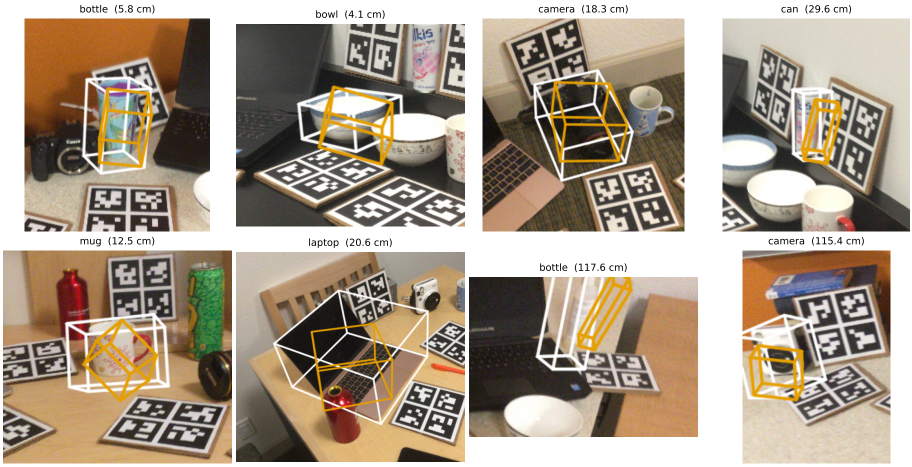

Fig. 1: Predicted metric 3D boxes (orange) vs. ground truth (white) on NOCS-Real275, RGB-only inference with the MoGe-2 anchor. Panel titles give the class and translation error. The first six panels span the six categories; the last two show the characteristic failure mode — distant, partially occluded cameras and bottles where the monocular anchor's depth error displaces the box along the viewing ray.

In-domain anchor caveat. ARKitScenes appears in MoGe-2's training mixture (8.3% weight), so part of our ARKitScenes gain reflects the anchor's in-domain advantage. NOCS-Real275 and Objectron are *not* in MoGe-2's training set; results there are genuine out-of-domain transfer for the anchor.

Amodal ground truth vs. visible-region input. Roughly 28% of ARKitScenes objects are frame-clipped: the annotation is a full-object (amodal) box while the model sees only the visible mask. Category tails (shelves 78%, sink 61% MAPE) concentrate exactly there, inflating the ARKitScenes mean; the median is robust to this.

Flat-object proportions. Laptop is the lone weak NOCS class at mAP@0.25 (21.6 vs. 69–96 for other classes) — a proportion failure of the shape prior on thin, open objects that the anchor cannot fix, since only the (frozen) shape model controls proportions.

Evaluation scope. Held-out sets are modest (464 instances) and single-seed; mAP is NOCS-only, under the localisation-given-GT-instances protocol (perfect detection assumed). Variance estimates and detector-in-the-loop evaluation are future work.

5.6 Qualitative results

6 Conclusion

Metric scale in depth-conditioned image-to-3D generation is not missing — it is delegated. The generator predicts a dimensionless object–scene ratio, and the conditioning pointmap supplies the units; with a metric anchor, a frozen generator is already a metric reconstruction system to within 7% on clean data. Once stated, this mechanism organises everything else: cross-source dimension error collapses into the anchor's single isotropic factor; size supervision must attach to the anchored pathway, because absolute-size regression from scale-free latents fails to transfer; and the right role for learning is the *per-instance correction of the anchor's own bias*, which our targeted layout fine-tuning achieves — surpassing the raw ground-truth-anchor oracle on NOCS while using no ground-truth depth at training or inference. We believe the delegation lens applies to any generation pipeline that normalises its conditioning by a scene statistic, and that the anchor-conditions-and-supervises pattern (using the pointmap itself as a label-free metric target, as MoGe-2 does at the pixel level) is the natural next step toward metric 3D reconstruction that scales beyond 3D-box-annotated data.

Acknowledgements

TODO

References

1. Ahmadyan, A., Zhang, L., Ablavatski, A., Wei, J., Grundmann, M.: Objectron: A large scale dataset of object-centric videos in the wild with pose annotations. In: CVPR (2021)
2. Baruch, G., Chen, Z., Dehghan, A., Dimry, T., Feigin, Y., Fu, P., Gebauer, T., Joffe, B., Kurz, D., Schwartz, A., Shulman, E.: ARKitScenes: A diverse real-world dataset for 3d indoor scene understanding using mobile RGB-D data. In: NeurIPS (2021)
3. Bhat, S.F., Birkl, R., Wofk, D., Wonka, P., Müller, M.: ZoeDepth: Zero-shot transfer by combining relative and metric depth (2023)
4. Brazil, G., Kumar, A., Straub, J., Ravi, N., Johnson, J., Gkioxari, G.: Omni3D: A large benchmark and model for 3D object detection in the wild. In: CVPR (2023)
5. Hu, E.J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W.: LoRA: Low-rank adaptation of large language models. In: ICLR (2022)
6. Krishnan, A., Kundu, A., Maninis, K.K., Hays, J., Brown, M.: OmniNOCS: A unified NOCS dataset and model for 3D lifting of 2D objects (2024)
7. Roberts, M., Ramapuram, J., Ranjan, A., Kumar, A., Bautista, M.A., Paczan, N., Webb, R., Susskind, J.M.: Hypersim: A photorealistic synthetic dataset for holistic indoor scene understanding. In: ICCV (2021)
8. SAM 3D Team, Chen, X., Chu, F.J., Gleize, P., Liang, K.J., Sax, A., Tang, H., Wang, W., Guo, M., Hardin, T., Li, X., Lin, A., Liu, J., Ma, Z., Sagar, A., Song, B., Wang, X., Yang, J., Zhang, B., Dollár, P., Gkioxari, G., Feiszli, M., Malik, J.: SAM 3D: 3Dfy anything in images (2025)

9. Wang, H., Sridhar, S., Huang, J., Valentin, J., Song, S., Guibas, L.J.: Normalized object coordinate space for category-level 6d object pose and size estimation. In: CVPR (2019)
10. Wang, R., Xu, S., Dai, C., Xiang, J., Deng, Y., Tong, X., Yang, J.: MoGe: Unlocking accurate monocular geometry estimation for open-domain images with optimal training supervision (2024)
11. Wang, R., Xu, S., Dong, Y., Deng, Y., Xiang, J., Lv, Z., Sun, G., Tong, X., Yang, J.: MoGe-2: Accurate monocular geometry with metric scale and sharp details (2025)
12. Xiang, J., Chen, X., Xu, S., Wang, R., Lv, Z., Deng, Y., Zhu, H., Dong, Y., Zhao, H., Yuan, N.J., Yang, J.: Native and compact structured latents for 3D generation (2025)
13. Xiang, J., Lv, Z., Xu, S., Deng, Y., Wang, R., Zhang, B., Chen, D., Tong, X., Yang, J.: TRELLIS: Structured 3D latents for scalable and versatile 3D generation. In: CVPR (2025)
14. Yin, W., Zhang, C., Chen, H., Cai, Z., Yu, G., Wang, K., Chen, X., Shen, C.: Metric3D: Towards zero-shot metric 3D prediction from a single image. In: ICCV (2023)
15. Zhang, L., Rao, A., Agrawala, M.: Adding conditional control to text-to-image diffusion models. In: ICCV (2023)